The process of drawing a sample from a population is known as _________.

Select the correct option

Reload Math Equations

- ( ) Survey research
- ( ) Census
- ( ) None of these
- (•) Sampling

Click to Save Answer & Move to Next Question

Which is the R command for selecting a sample of size 10 from the population yp <- c(11,150, 121, 198, 12, 136, 14, 129, 17, 115, 186, 110, 121, 15, 14):

Select the correct option

Reload Math Equations

- S <- sample(yp,1)
- S <- sample(yp,25)
- S <- sample(yp,10)
- S <- sample(yp,5)

Click to Save Answer & Move to Next Question

Which of the following would generally require the largest sample size?

Select the correct option

Reload Math Equations

- ( ) Systematic sampling
- ( ) Simple random sampling
- (•) Cluster sampling
- ( ) Proportional stratified sampling

Click to Save Answer & Move to Next Question

if
$Var(\bar{y}_{sy}) < Var(\bar{y}_{srs})$

:

Select the correct option

Reload Math Equations

- $\rho_w = 0$
- $\rho_w = \frac{1}{(nk-1)}$
- $\rho_w > -\frac{1}{(nk-1)}$
- $\rho_w < -\frac{1}{(nk-1)}$

Click to Save Answer & Move to Next Question

We obtain the 10000 random sample of size 6 under SRSWOR using the following Population (111, 150, 121, 198, 112, 136, 114, 129, 117, 115, 186, 110, 121, 115, 114) which is the R command for repeating this procedure 150 times?

Select the correct option

Reload Math Equations

- for (i in 1,1500)
- for (i in 1:1500)
- for (i , 1:150)
- for (i in 1:150)

Click to Save Answer & Move to Next Question

A sample is a representative part of___________.

Select the correct option

Reload Math Equations

- [x] Population
- [ ] Self distribution

Click to Save Answer & Move to Next Question

In random sampling, the probability of selecting an item from the population is:

Select the correct option

Reload Math Equations

- [x] Un-decided
- [ ] Unknown
- [ ] Known
- [ ] One

Click to Save Answer & Move to Next Question

Bias in which few respondents responds to offered questionnaire is classified as:

Select the correct option

Reload Math Equations

- distributed error
- non-responsive bias
- responsive bias
- concerning error

Click to Save Answer & Move to Next Question

The cluster 125,115,129,134,111 can be defined in R-language as:

Select the correct option

Reload Math Equations

- ( ) Clu<-125,115,129,134,111
- ( ) Clu<-(125,115,129,134,111)
- (•) Clu<-c(125,115,129,134,111)
- ( ) Clu<-k(125,115,129,134,111)

Click to Save Answer & Move to Next Question

Which one of the following is not a type of non-probability sampling?

Select the correct option

Reload Math Equations

- ( ) Snowball sampling
- ( ) Quota sampling
- ( ) Convenience sampling
- (•) Systematic sampling

Click to Save Answer & Move to Next Question

In systematic sampling, value of k is classified as:

Select the correct option

Reload Math Equations

- multistage interval
- secondary stage interval
- sampling interval
- sub-stage interval

Click to Save Answer & Move to Next Question

The cluster population mean in R-language is defined as:

Select the correct option

Reload Math Equations

- pop.mean=sum(Yi)/(N*N)
- pop.mean<sum(Yi)/(N)
- pop.mean=sum(Yi)/(N*M)
- pop.mean=sum(Yi)/(N)

Click to Save Answer & Move to Next Question

Mrs. Trahan samples her class by selecting every third person on her class list. Which type of sampling method is this?

Select the correct option

- Stratified
- Simple
- Cluster
- Systematic

Click to Save Answer & Move to Next Question

When the sample size increases, everything else remaining the same, the width of a confidence interval for a population parameter will:

Select the correct option

Reload Math Equations

- ( ) increase
- ( ) remain unchanged
- ( ) sometimes increase and sometimes decrease
- (•) decrease

Click to Save Answer & Move to Next Question

Which of the following would usually require the smallest sample size because of its efficiency?

Select the correct option

- One stage cluster sampling
- Quota sampling
- Two stage cluster sampling
- Simple random sampling

Click to Save Answer & Move to Next Question

If

$$\sum_{i=1}^{4}\sum_{j=1}^{5} y_{ij} = 1700,\ M = 5 \text{ and } n = 4$$

then the weighted mean is:

Select the correct option

Reload Math Equations

- (selected) 85
- 80
- 82
- 90

Click to Save Answer & Move to Next Question

$$4\times5=20 \quad 1700\div20=85$$

When population is following the linear trend for n>1:

Select the correct option

Reload Math Equations

- $Var(\bar{y}_{str}) > Var(\bar{y}_{sy}) > Var(\bar{y}_{wor})$
- $Var(\bar{y}_{str}) < Var(\bar{y}_{sy}) < Var(\bar{y}_{wor})$
- None of above
- $Var(\bar{y}_{str}) = Var(\bar{y}_{sy}) = Var(\bar{y}_{wor})$

Click to Save Answer & Move to Next Question

For the following population x<-c(1,2,3,4,5), What will be the R command for finding mean?

Select the correct option

Reload Math Equations

- ( ) mean x
- (•) mean(x)
- ( ) mean{x}
- ( ) mean{c}

Click to Save Answer & Move to Next Question

A sample which is free from bias is called:

Select the correct option

Reload Math Equations

- Negatively biased
- Biased
- Positively biased
- Unbiased

Click to Save Answer & Move to Next Question

Which ONE of these sampling methods is a probability method?

Select the correct option

Reload Math Equations

- ( ) Judgment Sampling
- (•) Cluster Sampling
- ( ) Convenience Sampling
- ( ) Quota Sampling

Click to Save Answer & Move to Next Question

Another name of standard deviation of a sampling distribution is___________.

Select the correct option

Reload Math Equations

- [x] Standard error
- [ ] Replication error
- [ ] Sampling error
- [ ] Sample standard deviation

Click to Save Answer & Move to Next Question

If N= 200, N1=100; n1=15, the value of W1 is:

$$100 \div 200 = 0.5$$
$$n1 = N1 \div N$$

Select the correct option

○	0.44
○	0.21
○	0.36
●	0.50

Click to Save Answer & Move to Next Question

Q. How sampling is done by using R?

Ans: Sampling With replacement
Sample (y, size, replace=TRUE)

Sampling With replacement
Sample (y, size, replace=FALSE)

OR

Sample(y, size)

Short Question 3Marks

Q. Derive t[illegible]e of Sample Mean in Stratified Sampling using Proportional Allocation?

Proportional allocation example

Consider a population of size 700 consisting on three strata such that N_1=100, N_2=250 and N_3=350. The required sample size is 18. First we will allocate the sample size to each stratum according to proportional allocation.

Sample Size Allocation

N_1=100, N_2=250, N_3=350, N=700, n=18

This example is the same as a type of paper but values are different

5 marks

$$n_h = \frac{n}{N} N_h$$

$$n_1 = \frac{n}{N} N_1 = \frac{18}{700} 100 = 3$$

$$n_2 = \frac{n}{N} N_2 = \frac{18}{700} 250 = 6$$

$$n_3 = \frac{n}{N} N_3 = \frac{18}{700} 350 = 9$$

STA632 - Sampling Techniques (Quiz 2) Quiz Start Time:

Question # 1 of 10 (Start time: 02:29:55 PM, 03 February 2023)

The product estimator for population mean is defined as:

Select the correct option

○	None of the above
○	$\bar{y}_2 = \frac{\bar{y}}{\bar{X}}$
○	$\bar{y}_1 = \frac{\bar{y}\bar{x}}{\bar{X}}$ @
○	$\bar{y}_3 = \frac{\bar{y}}{\bar{x}}\bar{X}$

Click to Save Answer

STA632 - Sampling Techniques (Quiz 2)

Question # 6 of 10 (Start time: 02:36:25 PM, 03 February 2023)

What is the bias of ratio estimator in two stage sampling?

Select the correct option

- $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2(V_{02} - V_{20} + 2V_{11})$
- $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2(V_{02} - V_{20} - 2V_{11})$
- None of the above
- $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2(V_{02} + V_{20} - 2V_{11})$

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STA632 - Sampling Techniques (Quiz 2)

Question # 7 of 10 (Start time: 02:37:18 PM, 03 February 2023)

Farmer Joe separates his farm into 10 regions. He then randomly selected 5 trees from each re
produced on his apple tree farm. This is _____________ sampling.

Select the correct option

- Systematic
- Cluster
- Stratified
- Simple

STA632 - Sampling Techniques (Quiz 2)

Quiz Start Time: 02:29 PM, 03 February 2023

Question # 7 of 10 (Start time: 02:37:18 PM, 03 February 2023)

Total Marks: 1

Farmer Joe separates his farm into 10 regions. He then randomly selected 5 trees from each region to estimate the number of apples produced on his apple tree farm. This is _____________ sampling.

Select the correct option

- Systematic
- Cluster
- Stratified
- Simple

Click to Save Answer & Move to Next Question

STA632 - Sampling Techniques (Quiz 2)

Question # 8 of 10 (Start time: 02:38:49 PM, 03 February 2023)

Which is the formula of separate type ratio estimator (using 2 strata)?

Select the correct option

- $\bar{y}_{rsst} = \sum_{h=1}^{2} W_h \bar{y}_{rh}$
- $\bar{y}_{rsst} = \sum_{h=1}^{4} W_h \bar{y}_{rh}$
- $\bar{y}_{rsst} = \sum_{h=1}^{3} W_h \bar{y}_{rh}$
- $\bar{y}_{rsst} = \sum_{h=1}^{5} W_h \bar{y}_{rh}$

2:41 PM

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Question # 10 of 10 (Start time: 02:40:52 PM, 03 February 2023)

What is the formula of product estimator?

Select the correct option

- ○ $\frac{\bar{y}\bar{x}}{\bar{X}}$
- ○ $\frac{\bar{X}\bar{x}}{\bar{y}}$
- ○ $\frac{\bar{X}\bar{y}}{\bar{x}}$
- ○ None of the above

12:24 PM 4G 9.1 K/s 67

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Question # 1 of 10 (Start time: 08:48:43 AM, 02 February 2023)

Which of the following is NOT a form of non-probability sampling?

Select the correct option

- Quota sampling
- Stratified sampling
- Snowball sampling
- Purposive sampling

Click to Save Ans

past accor

* Ranked set sampling was introduced by McIntyre in the year $\boxed{(1952)}$
* The list of all units in a population is called $\boxed{\text{Sampling frame}}$

(23) 92 module

* Which of the following sampling methods is best way to select a group of people for a study if you are interested in $\boxed{\text{random sampling}}$
* ________ is a set of elements taken from a larger population according to certain rules $\boxed{\text{Sample}}$

96 module

* Auxiliary variable is called $\boxed{\text{supporting varia}}$
* Grant estimated the population of England in which year $\boxed{1662}$

100 module

* Hartley Ross proposed the unbiased regression estimator in which year $\boxed{1954}$
* When Hertley-Ross proposed the unbiased ration estimator. $\boxed{1954}$

104 module

* What is the R command to find out the mean of x $\boxed{\text{mean(x)}}$
* What is the R command to find out the variance of x $\boxed{\text{var(x)}}$

$\boxed{28.28}$

107 module

★ If sample size is large sample mean to approaches to population mean

$\boxed{\text{True}}$

★ Simple random sampling and systematic
★ Sampling are $\boxed{\text{Probability Sampling}}$

The estimate of ratio in ratio method of estimation is consistent of estimate of Population ratio.

$\boxed{\text{True}}$

Module 131 Double sampling is also known as $\boxed{\text{two phase Sampling}}$

★ What is the R command for writing variance-covariance matrix $\Sigma = \begin{bmatrix} 1 & .75 \\ .75 & 1 \end{bmatrix}$

$\boxed{\text{Sig = matrix(c(1, 0.75, 0.75, 1), ncol = 2))}}$

12:24 PM

STA632 – Sampling Techniques (Quiz 2)

Question # 1 of 10 (Start time: 01:54:51 PM, 03 February 2023)

What is the mean of ratio estimator in two stage sampling?

Select the correct option

- ○ $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2(V_{02} - V_{20} - 2V_{11})$
- @ $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2(V_{02} + V_{20} - 2V_{11})$
- ○ None of the above
- ○ $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2(V_{02} - V_{20} + 2V_{11})$

12:23 PM

STA632 - Sampling Techniques (Quiz 2)

Question # 3 of 10 (Start time: 01:58:13 PM, 03 February 2023)

The expression of combined type ratio estimator is:

Select the correct option

- $\bar{y}_{rst} = \frac{\bar{y}_{st}}{\bar{X}} \bar{x}_{st}$
- None of the above
- $\bar{y}_{rst} = \frac{\bar{y}_{st}}{\bar{x}_{st}} \bar{X}$
- $\bar{y}_{rst} = \frac{\bar{Y}_{st}}{\bar{x}_{st}} \bar{X}$

12:23 PM

STA632 - Sampling Techniques (Quiz 2)

Quiz S

Question # 4 of 10 (Start time: 01:59:24 PM, 03 February 2023)

What is the R command for writing variance-covariance matrix

$$\Sigma = \begin{bmatrix} 1 & 0.65 \\ 0.65 & 1 \end{bmatrix}$$

?

Select the correct option

- None of the above
- sig=matrix(c(1,0.65,0.65,1),ncol=4)
- @ sig=matrix(c(1,0.65,0.65,1),ncol=2)
- sig=matrix(c(1,1,0.65,0.65),ncol=2)

Click to Save

STA632 - Sampling Techniques (Quiz 2)

Question # 4 of 10 (Start time: 01:59:24 PM, 03 February 2023)

What is the R command for writing variance-covariance matrix

?

Select the correct option

- None of the above
- sig=matrix(c(1,0.65,0.65,1),ncol=4)
- a sig=matrix(c(1,0.65,0.65,1),ncol=2)
- sig=matrix(c(1,1,0.65,0.65),ncol=2);

STA632 - Sampling Techniques (Quiz 2)

Question # 5 of 10 (Start time: 02:00:09 PM, 03 February 2023)

What is the bias of ratio estimator in two stage sampling?

Select the correct option

- $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2(V_{02} - V_{20} + 2V_{11})$
- $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2(V_{02} - V_{20} - 2V_{11})$
- @ $MSE(\bar{y}_{r2s}) \approx \bar{Y}^2(V_{02} + V_{20} - 2V_{11})$
- None of the above

STA632 - Sampling Techniques (Quiz 2)

Question # 6 of 10 (Start time: 02:01:24 PM, 03 February 2023)

Which is the formula of separate type ratio estimator (using 4 strata)?

Select the correct option

- $${\bar y_{rsst}} = \sum\limits_{h = 1}^6 {{W_h}{{\bar y}_{rh}}}$$
- $${\bar y_{rsst}} = \sum\limits_{h = 1}^5 {{W_h}{{\bar y}_{rh}}}$$
- $${\bar y_{rsst}} = \sum\limits_{h = 1}^4 {{W_h}{{\bar y}_{rh}}}$$
- $${\bar y_{rsst}} = \sum\limits_{h = 1}^3 {{W_h}{{\bar y}_{rh}}}$$

12:23 PM

STA632 - Sampling Techniques (Quiz 2)

Question # 8 of 10 (Start time: 02:02:39 PM, 03 February 2023)

What is MSE of product estimator in double sampling?

Select the correct option

- None of the above
- $MSE(\bar{y}_{rd}) = \bar{X}^2[\lambda_2 C_y^2 + (\lambda_2 - \lambda_1)(C_x^2 + 2C_x C_y \rho xy)]$
- $MSE(\bar{y}_{rd}) = \bar{Y}^2[\lambda_2 C_x^2 + (\lambda_2 - \lambda_1)(C_y^2 + 2C_x C_y \rho xy)]$
- $MSE(\bar{y}_{rd}) = \bar{Y}^2[\lambda_2 C_y^2 + (\lambda_2 - \lambda_1)(C_x^2 + 2C_x C_y \rho xy)]$

12:23 PM

Question # 10 of 10 (Start time: 02:05:47 PM, 03 February [illegible])

Which is the formula of separate type ratio estimator (using 4 strata)?

Select the correct option

- $${{\bar y_{rsst}} = \sum\limits_{h = 1}^6 {{W_h}{{\bar y}_{rh}}}}$$
- $${{\bar y_{rsst}} = \sum\limits_{h = 1}^5 {{W_h}{{\bar y}_{rh}}}}$$
- $${{\bar y_{rsst}} = \sum\limits_{h = 1}^3 {{W_h}{{\bar y}_{rh}}}}$$
- $${{\bar y_{rsst}} = \sum\limits_{h = 1}^4 {{W_h}{{\bar y}_{rh}}}}$$

The product estimator for population mean is defined as:

Select the correct option

- $\bar{y}_3 = \frac{\bar{y}}{\bar{x}}\bar{X}$
- $\bar{y}_1 = \frac{\bar{y}\bar{x}}{\bar{X}}$ ✓
- $\bar{y}_2 = \frac{\bar{y}}{\bar{X}}$
- None of the above

12:22 PM

Question # 9 of 10 (Start time: 03:01:11 PM, 03 February 2023)

If N= 4500, NI=1000; n1=150, N2=2000; n1=15; n2=15, the value of W2 is:

Select the correct option

- 0.44
- 0.40
- 0.50